Chapter 2 exercises

DS02-01 • CSR and transpose

A 3×4 matrix has entries `(0,1,5),(1,0,-2),(1,3,7),(2,1,4)`. Give all three CSR arrays. Use counts and prefix sums to construct the transposed COO, listing every output position. Transpose again to check the result.

Hint. Keep immutable starting positions separate from mutable write cursors.

DS02-02 • Four-dimensional addresses and capacity

A row-major array has dimensions $D_0 \times D_1 \times D_2 \times D_3$, zero-based indices, and w bytes per element. Derive the element and byte offsets of (i_0, i_1, i_2, i_3). State the bounds to check before allocation.

Hint. Interpret successive dimensions as mixed-radix digits.

DS02-03 • KMP trace

Compute the prefix function of `P=aaaaab`. Find its first match in `T=aaaaaaaaab`, listing q updates after mismatches. Compare the repeated work of naive matching on longer texts of the same form.

Hint. Identify reusable prefixes and suffixes inside the matched segment.

DS02-04 • Does CSR always save space?

Use 4-byte indices and 8-byte values. Ignore container headers and matrix dimensions. For R rows and k nonzeros, derive COO and CSR storage in bytes and the condition under which CSR is smaller. Interpret `R=1,000,000,k=100`.

References and attribution

Ellis Horowitz, Sartaj Sahni, Dinesh Mehta. Fundamentals of Data Structures in C++, 2nd Edition.

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https://github.com/xinhi6716/computer-science-self-study